

Relative atomic mass and relative formula mass

Learning outcomes

By the end of this section you should be able to:

- Explain that all atoms have a mass that is called relative atomic mass (A_r) and is measured in comparison to the carbon-12 isotope.
- Determine the relative formula masses (M_r) from relative atomic masses (A_r) using the periodic table.

When you buy fruits and vegetables at a store or market, you are often charged based on how heavy the item is, so that you pay more money for a larger quantity. Have you ever wondered about the units of measurement that we use to determine how heavy an object is? Worldwide, there are different systems of measurement, therefore scientists use an agreed system of measurement developed by The International System of Units, abbreviated as SI.

Nature of Science

Aspect: Measurement

SI uses units of kilograms to measure the mass of an object, but where did this unit come from? SI originally used a cylinder made of an alloy of platinum and iridium that was designated as the mass of 1 kilogram and other objects were measured relative to this standard (**Figure 1**). Therefore, if an object has twice the mass of this cylinder, it is said to have a mass of 2 kilograms. If an object has half the mass of this standard, it is said to have a mass of 0.5 kilograms, which can also be expressed as 500 grams, using metric conversions.



Figure 1. The International System of Units has a reference standard of a 1 kilogram cylinder. All other objects were measured relative to this standard.

Source: "International prototype of the kilogram aka Le Grand K

(https://commons.wikimedia.org/wiki/File:International_prototype_of_the_kilogram_aka_Le_Grand_K.jpg)"

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The definition of this unit has been recently revised to be based on the Planck constant of $h = 6.63 \times 10^{-34} \text{ J s}$, which is listed in [section 2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/>) of the DP chemistry data booklet.

Chemical entities have very small masses, which would not register on a balance, so we also use a reference standard to express the masses of elements and compounds.

Recall from [section S1.2.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>) that the particles that contribute to an atom's mass are the nuclear particles: the protons and neutrons. Protons and neutrons have almost the same mass, which is extremely small and cannot be measured directly with a balance, so all atoms are measured relative to a reference element, known as relative mass.

Chemists have chosen the carbon-12 isotope, as it contains equal numbers of protons and neutrons. The mass of the ^{12}C is assigned a value of 12, and the masses of all other atoms are compared relative to ^{12}C . For instance, an atom of the hydrogen-1 isotope, containing only one proton, is $\frac{1}{12}$ the mass of ^{12}C (**Figure 2**).

Concept

The current definition of the mole was announced in 2018 and is recommended by the IUPAC (<https://iupac.org/new-definition-mole-arrived/#:~:text=One%20mole%20contains%20exactly%206.022,is%20called%20the%20Avogadro%2>). The mole was previously defined as the amount of substance containing the same number of particles as the number of atoms contained in exactly 12 grams of the isotope carbon-12 (see [section S1.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/>)).

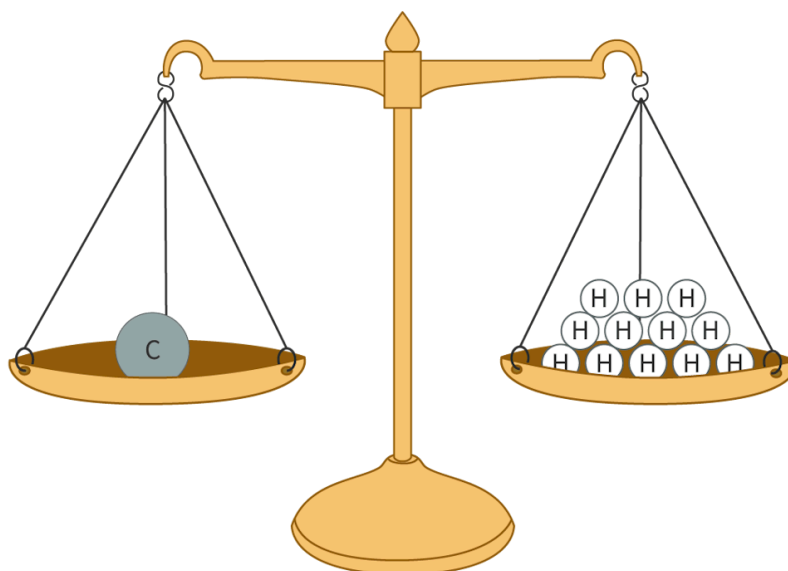


Figure 2. Hydrogen atoms are $\frac{1}{12}$ the mass of one carbon atom.

 More information for figure 2

The image depicts a balance scale. On the left side of the scale, there is a single sphere labeled with a "C," representing a carbon atom. On the right side, there are twelve smaller spheres, each labeled with an "H," representing hydrogen atoms. This visual illustrates the comparison in mass between a single carbon atom and twelve hydrogen atoms, highlighting that twelve hydrogen atoms balance out the mass of one carbon atom, indicating the relative atomic mass relationship.

[Generated by AI]

In a given sample of atoms, there will be a variety of naturally occurring isotopes. Recall that you learned how to calculate the relative atomic mass for elements containing different isotopes in [section S1.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/>) and that these values are included in the periodic table. Since atomic masses are determined as a comparison, rather than by a measurement, these masses, known as relative atomic mass (A_r), do not have units.

Making connections

Atoms increase in mass as their position descends in the periodic table. What properties might be related to this trend? See [subtopic S3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>) to find the main patterns used to organise the elements in the periodic table.

Try the drag and drop in **Interactive 1** to match the relative atomic masses of these elements.

1.01	12.01	32.06	55.85
Carbon C	Iron Fe	Hydrogen H	Sulfur S
☒ Check			

Interactive 1. Identify the Relative Atomic Masses of These Elements.

Relative formula mass, M_r

Recall from [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) that pure substances can either be elements or compounds. Elements are represented by atoms, but compounds contain two or more atoms bonded together. You should know by now that atoms of different elements have a mass that is relative to carbon-12. But what happens when atoms combine? We need to add the masses of all the atoms in the formula to calculate the relative formula mass (M_r) for that compound.

The relative atomic mass of oxygen is 16.00, as given in the periodic table ([section 7](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) of the DP chemistry data booklet). Therefore, the relative formula mass of the element oxygen will be $16.00 + 16.00 = 32.00$, as we know its formula is O_2 . If the form of oxygen we are studying is ozone, then the formula is O_3 , and its relative formula mass will be $16.00 \times 3 = 48.00$ (once again, notice that neither relative atomic masses nor relative formula masses have units).

Note that the IB now uses the term relative formula mass to include both molecular compounds and ionic compounds. Previously, the term relative molecular mass was used exclusively for molecular compounds and relative formula mass for ionic compounds. In this text, the term relative formula mass is used for both types of compounds.

Practise with some worked examples.

Worked example 1

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of sodium chloride, NaCl.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question asks for the relative formula mass.
Step 2: write out the relative atomic masses for each element in the compound. The values can be found in the <u>periodic table</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{Na}): 22.99$ $A_r(\text{Cl}): 35.45$
Step 3: the M_r is the sum of the A_r of all the elements in the compound.	$M_r(\text{NaCl}) =$
Step 4: add the A_r values together.	$M_r(\text{NaCl}) =$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 58.44$

Worked example 2

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of carbon dioxide, CO₂:

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question relative form
Step 2: write out the relative atomic masses for each element in the compound. The values can be found in the <u>periodic table</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{C}): 12.01$ $A_r(\text{O}): 16.00$
Step 3: the M_r is the sum of the A_r of all the elements in the compound.	$M_r(\text{CO}_2) =$
Step 4: add the A_r values together.	$M_r(\text{CO}_2) =$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 44.01$

Worked example 3

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of the polyatomic carbonate ion, CO_3^{2-} .

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question in formula mass o
Step 2: write out the relative atomic masses for each element in the ion. The values can be found in the <u>periodic table</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{C}): 12.01$ $A_r(\text{O}): 16.00$
Step 3: the M_r is the sum of the A_r of all the elements in the ion.	$M_r(\text{CO}_3^{2-}) =$

Solution steps	Calculations
Step 4: add the A_r values together.	$M_r(\text{CO}_3^{2-}) =$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 60.01$

Note: Recall that the mass of an atom is mainly concentrated in its nuclear particles (protons and neutrons) so the extra electrons in carbonate will not change its mass significantly (neither it will the lack of electrons when the ion is positively charged).

Worked example 4

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of calcium phosphate, $\text{Ca}_3(\text{PO}_4)_2$.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding the relative formula mass of a compound.
Step 2: write out the relative atomic masses for each element in the compound. The values can be found in the periodic table (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{Ca}): 40.08$ $A_r(\text{P}): 30.97$ $A_r(\text{O}): 16.00$
Step 3: the M_r is the sum of the A_r of all the elements in the compound.	$M_r(\text{Ca}_3(\text{PO}_4)_2) = (3 \times A_r(\text{Ca})) +$

Solution steps	Calculations
Step 4: add the A_r values together.	$M_r(\text{Ca}_3(\text{PO}_4)_2) = (3 \times 40.08) +$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 310.18$

Note: when there are parentheses around a group of atoms, you must multiply the quantity of each atom by the number on the outside of the parentheses. Therefore, there are $(1 \times 2) = 2$ atoms of P and $(4 \times 2) = 8$ atoms of oxygen.

Use the [periodic table](#) to determine the relative atomic mass or relative formula mass for the following elements and compounds.

 48.00	 40.31	 2.02	 44.01
 36.46	 18.02	 32.00	 28.02
Carbon Dioxide CO₂	Water H₂O	Hydrochloric Acid HCl	Magnesium Oxide MgO
Oxygen O₂	Nitrogen N₂	Hydrogen H₂	Ozone O₃
☒ Check			

Interactive 2. Relative Atomic Masses of Some Elements.

Practical skills

Tool 1: Experimental techniques — Applying techniques

You might notice in the laboratory that some ionic compounds have H_2O in the chemical formula and are labelled as a hydrate. For example, copper(II) chloride dihydrate has the formula $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$. These are known as hydrates because they have a high attraction for water molecules. When using these substances in calculations, you must include the molar mass of the water molecules as well. For example, has a relative formula mass of:

$$A_r(\text{Cu}): 63.55 \quad (\text{Cl}): 35.45 \quad (\text{H}): 1.01 \quad (\text{O}): 16.00$$

$$M_r: 63.55 + (2 \times 35.45) + (2 \times 18.02) = 170.49$$

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Presenting data appropriately
- **Time required to complete activity:** 25—30 minutes
- **Activity type:** Individual activity

1. Copy and complete the table below using the [periodic table](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) in the DP chemistry data booklet.

Name of element	Symbol
hydrogen	
	He
nitrogen	
	O
sodium	
	Mg

Name of element	Symbol

2. Copy and complete the table below using the periodic table (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) in the DP chemistry data booklet.

Name of compound	Symbol
water	H ₂ O
methane	CH ₄
ammonia	NH ₃
caffeine	C ₈ H ₁₀ N ₄ O ₂
aluminium nitrate	Al(NO ₃) ₃

3. Check your answers with a classmate.